High school · counting

Paths, combinations and Pascal

Progressive exercises

Exercise 1

Find C(8,2) and explain why it equals C(8,6).

Exercise 2

Count monotone paths from (0,0) to (4,3).

Exercise 3

How many of these paths pass through (2,1)?

Exercise 4

Choose a team of three from eight people. Two specified people cannot be together. How many teams remain possible?